

Experiment 15

Inclined Plane

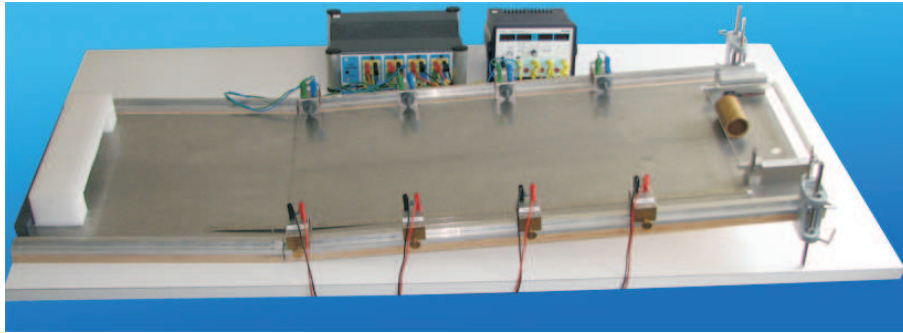


Figure 1: *Experimental setup.*

I Measurement setup

- Height-adjustable runway
- Light barrier with control unit
- Spirit level
- Ruler
- Rolling bodies
 - Solid cylinder (aluminum, $\rho = 2,70 \text{ g/cm}^3$)
 - Hollow cylinder (brass, $\rho = 8,44 \text{ g/cm}^3$)
 - Composite cylinder: sheath of aluminum, core of brass.
- Calliper
- Balance

II Literature

- W. Walcher, *Praktikum der Physik*, B.G.Teubner Stuttgart,
- Gerthsen, Bergmann-Schäfer, Tipler.
- <http://www.physi.uni-heidelberg.de/Einrichtungen/AP/>.

III Preparation

Prepare yourself for the following topics: Speed, acceleration, force, weight, sliding friction, rolling friction, moment of force, angular momentum, moment of inertia, momentum, momentum conservation, mechanical energy forms, energy conservation.

Comprehension questions:

1. Calculate the moment of inertia for the following bodies, where the rotating axis is the symmetry axis.
 - Solid cylinder
 - Hollow cylinder

Perform the calculation by integration from the general definition of the moment of inertia.
2. A „frictionless“ cuboid, a solid cylinder, a hollow cylinder and a solid ball, each with the same radius, slide or roll down an inclined plane. Compare the movements with each other. Which body comes first „below“ ?
3. A „frictional“ cuboid should be trimmed by surface treatment, that its acceleration on an inclined plane (10° inclination) is the same that the acceleration of a solid cylinder. To what value would you then have to set the sliding coefficient? (Hint: In addition to the downward gravity force, the sliding friction force is acting on the cuboid.)

IV Tasks

- Determine the acceleration on an inclined plane for two rolling bodies.
- Studies on the law of conservation of energy.

V Basics

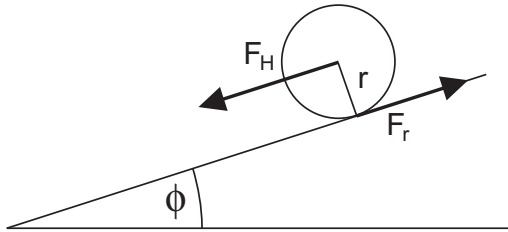


Figure 2: Derivation of the acceleration on the inclined plane.

The frictional force F_r (static friction force) of a rolling body on a „ramp“ causes at the radius r the moment of force

$$M = F_R \cdot r = I \frac{d\omega}{dt}. \quad (1)$$

Where ω is the angular velocity and I the moment of inertia of the rolling body. With the rolling condition $v_s = r\omega$, where the index „s“ refers to the center of gravity, this equation can be reshaped as:

$$M = F_R \cdot r = \frac{I}{r} \frac{dv_s}{dt} = \frac{I_s}{r} \cdot a_s \quad (2)$$

and thus

$$F_R = \frac{I}{r^2} \cdot a_s. \quad (3)$$

On the center of gravity of the rolling body acts the downhill force F_H and the frictional force in the opposite direction:

$$ma_s = mg \sin \varphi - F_R \quad (4)$$

The equations (3) and (4) allow the elimination of the friction force:

$$ma_s = mg \sin \varphi - \frac{I}{r^2} \cdot a_s \quad (5)$$

and you get for the acceleration of the center of gravity:

$$a_s = \frac{mg \sin \varphi}{m + \frac{I}{r^2}}. \quad (6)$$

VI Execution of the experiment

Familiarise yourself with time measurement technology first. The light barriers and the stopwatch are already wired. The stop inputs of the clock are triggered by the four light barriers. The light barriers each consist of a transmitter (infrared LED) and a receiver (infrared photodiode). Both are connected to a common channel of the light barrier control unit. The output of each channel of the control unit is connected to the corresponding stop input of the clock. The LEDs on the control unit indicate whether the light barriers are activated, that is, whether the photodiode is sufficiently illuminated from the opposite LED.

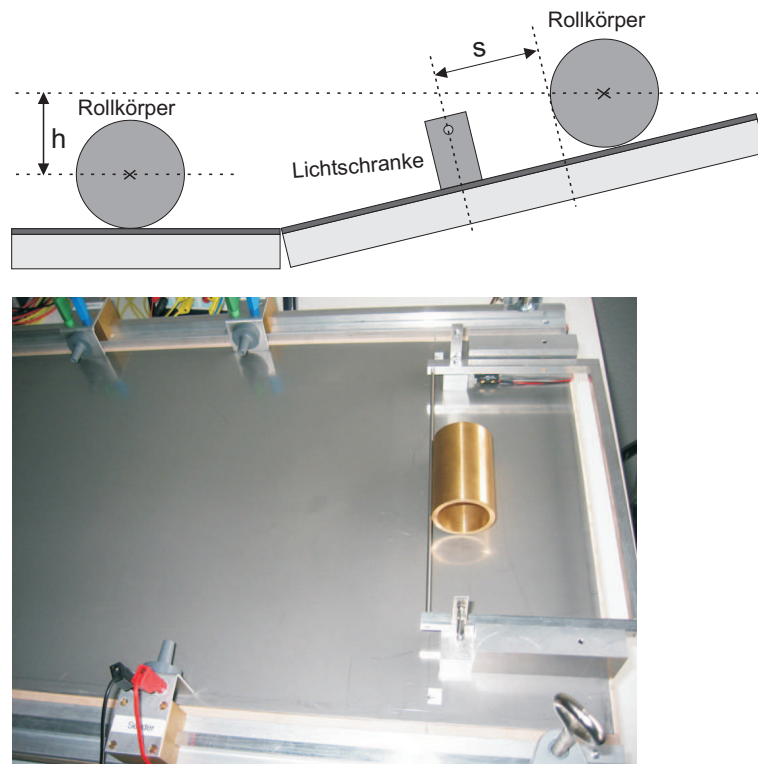
1. Sketch the experimental setup.

2. Measuring the inclined plane and the rolling bodies.

For the following studies on the inclined plane, the test bodies (full and hollow cylinder) must be measured accurately. Use a calliper to get the diameters.

Check with the spirit level whether the inclined plane has the same inclination over the entire width. If this is not the case, you must readjust the plate by adjusting the height of one of the „feet“. The Plate is very heavy! This is best done together with your partner. One holds the plate, the other loosens the two side screws and varies the length of the steel rod. Carefully think about which length and height you need to measure to calculate the inclination angle of the plane (Abbildung (3)).

3. Examination of the movement type at the inclined plane

Figure 3: *Sketch of the setup.*

First, check whether the rolling bodies are moving uniformly accelerated. For this, you should place the light barriers at **reasonable intervals** on the inclined plane ($t \propto \sqrt{s}$).

Examine qualitatively the accelerations of the rolling bodies. All three cylinders fit together side by side in the starting mechanism. Explain without calculation why the bodies need different lengths of time to arrive below.

Use only the solid and hollow cylinders for the following measurements and any inclination of the plane. The distances from the rolling body to the individual light barriers, must be measured from the center of the light entry openings at the light barriers to the point of contact of the rolling element at the start

mechanism. The starting line of the rolling body at the starting mechanism is marked on the stainless steel surface next to the starting mechanism.

4. Accurate quantitative analysis of acceleration

Determine the acceleration for two bodies (solid and hollow cylinders). Measure the distances s of the light barriers from the starting mechanism (see Figure 3). Stop 5 times the time the rolling bodies need to pass through the individual light barriers.

5. Studies on the law of energy conservation

The initial potential energy of the rolling body is converted into kinetic energy. The kinetic Energy can be divided into a rotation and a translation part, so that the total energy is given by the following equation:

$$W_{ges} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + mgh, \quad (7)$$

where v is the translation velocity, I the moment of inertia and ω the angular velocity of the rolling body. In this part of the experiment you measure the translational speed of the various rolling bodies on the horizontal section at the end of the inclined plane. For this you have to place two light barriers at the horizontal section. Perform the measurements 5 times each. The kinetic energy determined from the measured velocity is compared with the initial potential energy. To calculate the potential energy you need the height difference between start and end position of the rolling bodies.

VII Evaluation

to 4: Plot the distances s versus t^2 and calculate from the slope the acceleration. Compare this to the value deriving from the mass and geometry of the rolling body as well as from the inclination angle of the inclined plane.

to 5: Calculate the kinetic energy (translation and rotation energy) and compare this with the potential energy. Are the results compatible within the statistical errors? Which systematic errors could exist?